MIL-T-4220B(USAF)
3 March 1959
Superseding
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12 June 1953

MILITARY SPECIFICATION

TEST OSCILLATOR SET AN/PRM-10( )

1. SCOPE

1.1 This specification covers one type of test oscillator set, designated Test Oscillator Set AN/PRM-10( ), hereinafter referred to as the "oscillator".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-R-3065	Rubber and Synthetic Rubber Compounds, General Purpose (Except Tires, Inner Tubes, Sponge Rubber and Hard Rubber)
MIL-C-3767	Connectors, Electrical (Power, Bladed Type)
MIL-C-4003	Cement, General Purpose Synthetic Base
MIL-T-4807	Tests; Vibration and Shock, Ground Electronic Equipment (General Requirements For)
MIL-D-5028	Drawing and Data Lists; Preparation Of Manufacturer's (For Production Aeronautical and Associated Equipment)
MIL-T-9107	Test Reports; Preparation Of
MIL-F-14072	Finishes For Ground Signal Equipment
MIL-P-17555	Preparation For Delivery Of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-21200	Test Equipment For Use With Electronic and Fire Control Systems, General Specification For

STANDARDS

Federal

FED-STD-595	Colors

FSC 6625

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Military

MIL-STD-130 *Identification Marking of U.S. Military Property*

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143 Specifications and Standards; Use Of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 COMPONENTS.- The oscillator shall consist of:

Item No.	Quantity	Description	Requirement
1	7 each	Oscillator Coil Assembly	3.7.1
2	1 each	Tuning Head	3.7.2
3	1 each	Power Supply	3.7.3
4	1 each	Combination Case	3.7.4

3.3 STANDARD SAMPLE.- A standard sample oscillator will be furnished to demonstrate the minimum acceptable standard of materials, design, workmanship, and performance not specified herein. The standard sample will also be used to determine compliance with the required functional and dimensional interchangeability. If the standard sample conflicts with the requirements of this specification, this specification shall govern.

3.4 GENERAL SPECIFICATION.- The requirements of MIL-T-21200 apply as requirements of this specification. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern. Exceptions and additions to the general specification shall be as follows.

3.4.1 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services, not specified herein, shall be selected in accordance with ANA Bulletin No. 143.

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3.4.2 INTERCHANGEABILITY.- In addition to the requirements of MIL-T-21200, all parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-5028.

3.4.3 MECHANICAL.- The oscillator shall be designed and constructed to meet the vibration conditions as detailed in Section 4 of this specification.

3.4.4 MARKING AND IDENTIFICATION.- Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130. The following special markings shall be added:

a. The nomenclature "AN/PRM-10( )" shall be stenciled on the top and two adjacent sides of the combination case.

b. The information "115VAC 50-1000 cycles" shall be added to the oscillator nameplate.

3.4.5 CLASSIFICATION.- The oscillator shall be Class 2A.

3.5 DESIGN AND CONSTRUCTION.- The oscillator shall consist of an oscillator circuit and a regulated power supply. The oscillator circuit (contained in the tuning head) shall be capable of producing a continuous or modulated radio-frequency (r-f) signal or performing as a non-oscillating detector. The r-f signal shall be amplitude modulated at will by either an internal or external audio-frequency (a-f) signal. The oscillator circuit shall be tuned by a variable capacitor matched with any one of a set of seven plug-in type coils. The coils shall be physically extended and exposed to permit coupling to the circuit under test. A phone jack shall be provided to allow insertion of a high impedance telephone headset into the circuit to permit aural recognition of indications. Relative coupling and resonance shall be indicated by a meter electrically located in the grid circuit of the oscillator.

3.6 PERFORMANCE.- The oscillator shall be capable of being used as follows:

a. R-f signal generator: The oscillator shall provide from 2 to 400 megacycles (mc) r-f signals for general test purposes.

b. Oscillating frequency meter (grid dip meter): When magnetically coupled to an unenergized r-f circuit in the range of 2 to 400 mc, the oscillator shall visually indicate the resonant frequency by a sharp dip in the indicating meter.

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c. *Oscillating detector:* The oscillator shall audibly indicate by beat notes or clicks in high impedance headphones the fundamental and harmonic frequencies of energized r-f circuits. The frequency shall be read on the frequency dial. (The headphones are not a part of this specification.)

d. Nonoscillating diode detector (absorbtion type meter): The oscillator shall visually and audibly indicate the presence of fundamental and harmonic frequencies in energized r-f circuits. Visual indication shall be a peak meter reading. Audible indication shall be in the headphones.

3.6.1 OPERATING CHARACTERISTICS.- The oscillator shall have the following operating characteristics:

a. Frequency range: 2 to 400 mc.

b. Signals produced: Continuous wave (cw) or modulated cw. The signals shall be at least 30 percent amplitude modulated by an internal 1000-cycle oscillator circuit or a 50 to 10,000-cycle external source over the entire frequency range. Harmonic content shall be at least 20 decibels (db) below the fundamental frequency amplitude.

c. Accuracy of frequency indications: Within plus or minus 1.5 percent for any frequency from 2 to 400 mc. This 1.5 percent accuracy shall be accomplished without the use of adjusting trimmers in the oscillatory circuit.

d. Hum level: Residual hum or hum modulation from any source in the oscillator shall not exceed 2 percent of the fundamental frequency amplitude.

3.7 *DETAILS OF COMPONENTS*

3.7.1 OSCILLATOR COIL ASSEMBLY.- Each coil assembly shall be of the plug-in type and shall have mechanical protection and high voltage electrical insulation. Coils shall be wound on an elongated base to facilitate probing. One coil assembly shall be provided for each of the following frequency bands:

a. 2 to 5 mc.
b. 5 to 10 mc.
c. 10 to 22 mc.
d. 22 to 45 mc.
e. 45 to 100 mc.
f. 100 to 250 mc.
g. 250 to 400 mc.

Frequency bands shall overlap by at least 10 percent.

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3.7.2 TUNING HEAD.- The tuning head shall contain the r-f oscillating and indicating facilities of the oscillator. The tuning head shall contain a socket to permit the plug-in of any one of the coil assemblies, an indicating meter, and a frequency dial. The tuner shall be enclosed in an aluminum alloy instrument case which shall have rounded edges. The tuning head shall be permanently attached to the power supply (see 3.7.3) by an electrical cord.

3.7.2.1 METER.- A ruggedized, sealed microammeter type MR-26-W-100 DC Microamperes shall be connected in the grid circuit of the oscillator to visually indicate points of resonance and the presence of fundamental and harmonic frequencies. Meter indications, while traversing the frequency band of each coil assembly, shall vary as specified in 4.3.1.2.4, Table I. Meter deflections shall be steady with no inherent dips or peaks when the tuning head is not coupled to a tuned or oscillating circuit. A meter sensitivity control which shall be controlled by a front panel knob shall be provided to enable strong and weak signals to be read on the same scale. The meter shall indicate at least 100 microamperes (ua) at any frequency with the sensitivity control at 2/3 of its limit of rotation.

3.7.2.2 OSCILLATOR TUNING.- A variable capacitor and a variable resistor both controlled by a common front panel knob shall be provided for tuning the r-f oscillator circuit. When the oscillator circuit is tuned, it shall be possible to read the tuned frequency on the frequency dial.

3.7.2.3 FREQUENCY DIAL.- A dial shall be provided to indicate actual frequency readings. The dial shall be marked in seven scales corresponding to the frequency bands (and overlap) of the seven oscillator coil assemblies.

3.7.3 POWER SUPPLY.- The power supply shall operate from 115-volt plus or minus 10 percent, 50 to 1000-cycle, single-phase, alternating-current (a-c) power and shall supply regulated voltages to the oscillator circuits. The power supply shall contain a 1000-cycle oscillator circuit to permit internal modulation of the r-f signal. It shall be possible to remove the power supply from the combination case for servicing.

3.7.3.1 POWER SUPPLY CHASSIS.- The power supply chassis shall contain dummy sockets to hold the oscillator coil assemblies when not in use and shall mount the following controls:

a. Power on-off toggle switch.

b. Power on indicator light.

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c. Fuse and spare fuse (0.3 amperes each).

d. Oscillator-diode toggle switch to enable the oscillator to be used as an r-f oscillator or a diode detector.

e. Modulated-cw toggle switch to enable the oscillator signal to be modulated or unmodulated (only when oscillator-diode switch is in oscillator position).

f. Jack to permit modulation by an external modulator (only when modulated-cw switch is in modulated position).

g. Jack to permit the insertion of headphones into the circuit for aural monitoring.

3.7.3.2 POWER CABLE.- The input power cable shall be 6 feet long with one end permanently attached to the power supply and the other end terminated in a type UP121M electrical plug, in accordance with MIL-C-3767.

3.7.4 COMBINATION CASE.- A watertight aluminum alloy combination case (MIL-T-21200) shall be provided. The volume of the case shall not exceed 525 cubic inches. The weight of the case when loaded shall not exceed 13 pounds. With the case latched, there shall be no damaging pressure on the coils or tuning head due to the position of cables in the case. A gasket shall be provided in the cover of the case. The gasket shall be made of material in accordance with MIL-R-3065, grade RS-300BK, and shall be secured to the cover by cement in accordance with MIL-C-4003. Spring-loaded latches, with latches on the case and hooks on the cover, shall be provided.

3.8 DIMENSIONS.- The oscillator shall fit into the 525 cubic inch (or smaller) combination case without damage.

3.9 WEIGHT.- The oscillator (and combination case) shall not weigh more than 13 pounds.

3.10 COLOR.- The oscillator shall be painted gloss gray, color No. 16376, in accordance with FED-STD-595.

3.11 FINISHES.- All finishes shall be in accordance with MIL-F-14072. The following items shall be rhodium flashed, over silver plate, according to the best commercial practice:

a. Rotor contacts on tuning capacitor.
b. Male and female contact plugs on the coil and tuning head, respectively.
c. Grounding strips in tuning head.
d. Capacitor split-gear.

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3.12 GOVERNMENT-LOANED PROPERTY.- When the contract or purchase order so provides, the Government will loan the following to the contractor upon his request:

Item No.	Stock No.	Part No.	Description	Quantity
1			Test Oscillator Set AN/PRM-10( )	1

4. QUALITY ASSURANCE PROVISIONS

4.1 GENERAL.- The sampling and inspection procedures for the oscillator shall be in accordance with MIL-T-21200 with the following additions and exceptions.

4.2 PREPRODUCTION TESTS

4.2.1 PREPRODUCTION TEST SAMPLES.- The preproduction test samples shall consist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under supervision of the procuring activity.

4.2.2 PREPRODUCTION TEST REPORT.- After completion of the preproduction tests by the contractor, he shall prepare a preproduction test report in accordance with MIL-T-9107 and furnish three complete copies of the report to the procuring activity

4.2.3 DESIGN DATA.- Design data will not be required except for three copies of a report by the contractor of all tests previously performed on the preproduction test samples.

4.2.4 PRESUBMISSIONING.- Presubmissioning will not be required.

4.2.5 HUMIDITY TEST.- The unsealed oscillator shall be subjected to 10 of the 48-hour humidity cycles with the combination case cover removed.

4.2.6 MECHANICAL TESTS.- The oscillator shall be subjected to the vibration test, Method I, in accordance with MIL-T-4807.

4.2.7 WATERTIGHT TEST.- The oscillator in the latched combination case shall be submerged to a depth of at least 3 feet in water at ambient temperature for a period of 4 hours. There shall be no evidence of moisture penetration.

4.2.8 COIL TEST.- One set of coils for the oscillator shall be selected at random from each lot of 50 or fraction thereof on the order and subjected to the following test. The protective

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covering of the coils shall be removed and the relative Q of each coil shall be measured and recorded. The coils shall be submerged to a depth of at least 6 inches in water at 120°F for a period of 4 hours. The coils shall be removed from the water and the Q measured. The Q shall not vary more than 10 percent during this test.

4.2.9 LIFE TEST.- The oscillator shall be operated continuously for 200 hours. During this test, the voltages, currents, loading, and method of excitation shall, insofar as is practical, correspond to actual field conditions. Upon completion of this test, the performance of the oscillator shall not be below the minimum requirements specified herein.

4.3 ACCEPTANCE TESTS

4.3.1 INDIVIDUAL TEST.- Each oscillator shall be subjected to the following tests:

a. Examination of product.
b. Functional tests.

In addition, each oscillator shall be subjected to any other tests specified herein which the procuring activity considers necessary to determine compliance with the requirements of this specification.

4.3.1.1 EXAMINATION OF PRODUCT.- Each oscillator shall be given a thorough mechanical and visual inspection and test to determine that the quality of material and workmanship is in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Completeness.
b. Finishes.
c. Identification markings and labels.
d. Wiring and soldering.
e. Loose fastening and securing devices or parts.
f. Operation of controls.
g. Dial and meter markings.
h. Fit of coils and tubes in sockets.
i. Accessibility of components and parts for servicing.
j. Weight.
k. Dimensions.

4.3.1.2 FUNCTIONAL TESTS.- Each oscillator and each individual component thereof shall be given a thorough electrical performance test to determine that all circuits are inherently sound and that overall performance of the equipment is in compliance with this specification. Tolerance limits shall be checked to determine

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that circuit elements have been selected in accordance with the requirements of each application. Electrical tests shall include, but not necessarily be limited to, the following tests.

4.3.1.2.1 FREQUENCY LIMITS, ACCURACY, AND OVERLAP TEST.- Each oscillator coil assembly shall be tested to determine if it meets the minimum requirements for frequency limits and overlap specified herein. The accuracy of the oscillator frequency indications shall be measured at two separate frequencies in each band. The accuracy shall be within plus or minus 1.5 percent.

4.3.1.2.2 HUM LEVEL TEST.- The a-c voltage at the B+ terminal of the power supply shall not exceed 3.4 volts root-mean-square when measured with a vacuum tube voltmeter.

4.3.1.2.3 SENSITIVITY CONTROL TEST

a. Set the sensitivity control at 2/3 maximum rotation.

b. Traverse the frequency band of each coil assembly.

c. The indicating meter shall read at least 100 ua at all times during this test.

4.3.1.2.4 METER INDICATION TEST

a. Traverse the frequency band of each coil assembly. At maximum meter reading, adjust meter to maximum reading as specified in Table I.

b. Retraverse the coil assembly frequency band.

c. The meter deflection shall vary between the maximum and minimum values specified in Table I.

d. There shall be no sharp deflections (dip or peaks) exceeding 3 ua.

Table I

METER INDICATION

Oscillator Coil Frequency Band (mc)	Maximum Meter Reading (ua)	Minimum Meter Reading (ua)
2 to 5	100	60
5 to 10	100	60
10 to 22	100	60

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Oscillator Coil Frequency Band (mc)	Maximum Meter Reading (ua)	Minimum Meter Reading (ua)
22 to 45	100	60
45 to 100	100	60
100 to 250	99	33
250 to 400	98	42

4.3.1.2.5 INTERNAL MODULATION TEST

a. Set the oscillator to produce a modulated signal

b. Plug the vertical input of an oscilloscope into the oscillator phone jack.

c. Connect the horizontal input of the oscilloscope to an a-f test oscillator.

d. Vary the a-f test oscillator frequency until a fixed loop appears on this oscilloscope.

e. The a-f test oscillator frequency shall be 1000 plus or minus 100 cycles.

f. Remove the a-f test oscillator. Measure the percent modulation on the oscilloscope. The percent modulation shall be at least 30 percent.

4.3.1.2.6 EXTERNAL MODULATION

a. Set the oscillator to produce a modulated signal.

b. Plug an external a-f test oscillator into the external modulator jack of the oscillator and vary the a-f oscillator frequency from 50 to 10,000 cycles.

c. Measure the percent modulation on an oscilloscope. It shall be possible to modulate the oscillator signal at least 30 percent at any frequency in this range.

4.3.1.2.7 GRID DIP TEST

a. The test coils shall be production coils mounted in a jig and shunted by a 1.5 to 6 microfarad ceramic capacitor to tune each coil to the frequency indicated in the column under the heading of "Test Coil Resonant Frequency" in Table II.

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b. The separation between the top of the housings of the test coil and the coil under test shall be as indicated in the column under the heading "Separation of Coils" in Table II.

c. Set the oscillator to produce a cw signal.

d. Place the tuning head near a test coil and tune the oscillator to the resonant frequency of the test coil.

e. Remove the tuning head from the immediate vicinity of the test coil and adjust the meter to read 100 ua.

f. Again place the tuning head within the distance specified on the tuning coil.

g. The meter reading shall not exceed (pointer shall dip at least as low as) the value specified in Table II.

Table II

GRID DIP TEST

Oscillator Coil Frequency Band (mc)	Test Coil Resonant Frequency (mc)	Separation of Coils (inch + .050 - .000)	Maximum Meter Reading (ua)
2 to 5	4.8	0.500	86
5 to 10	10.0	0.590	72
10 to 22	21.0	0.725	60
22 to 45	40.0	1.000	75
45 to 100	95.0	0.920	70
100 to 250	200.0	1.000	75
250 to 400	300.0	0.125	56

4.3.1.2.8 DETECTOR TEST

a. Unit No. 1 is the test unit:

(1) Using any coil of Unit No. 1, tune to any frequency in its band.

(2) Set the oscillator-diode switch in the oscillator position.

(3) Set the modulated-cw switch in the modulated position.

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b. Unit No. 2 is the unit under test and placed into operation as follows:

(1) Plug high-impedance headphones into oscillator phone jack.

(2) Set the oscillator-diode switch in the diode position.

(3) Place coil of Unit No. 2 near coil of Unit No. 1.

(4) Tune Unit No. 2 to frequency of Unit No. 1.

(5) A 1000-cycle modulated signal shall be heard.

4.4 ORDER OF TESTS.- The order in which tests are made shall be at the discretion of the procuring activity. Test results, regardless of the order in which tests are performed, shall in no case fall below the minimum requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 The oscillator shall be prepared and marked for shipment in accordance with MIL-P-17555. The shipment marking nomenclature shall be as follows: "Test Oscillator Set AN/PRM-10( )".

6. NOTES

6.1 INTENDED USE.- The oscillator is intended for use as an r-f signal generator and frequency meter in the range of 2 to 400 mc for testing various radio and radar equipments.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Conditions under which Government-loaned property will be made available to the contractor. (See 3.12.)

c. Number of preproduction test samples required. (See 4.2.1.)

d. Location and inspection of preproduction tests. (See 4.2.1.)

e. Applicable levels of preservation, packing, and packaging. (See 5.1.)

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6.3 NOMENCLATURE.- The parentheses, appearing in the nomenclature will be dropped or replaced by a letter to designate the particular design. Nomenclature for the individual components, where required, will be furnished upon application to the procuring activity.

6.4 DRAWINGS.- Upon specific request of the contractor to the procuring activity, a copy of the applicable drawings, if available, may be furnished for informational purposes only.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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RCSSES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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